National Institute of Open Schooling (NIOS)
Senior Secondary Course
Lesson – 27: Differentiation of Trigonometric Functions
Worksheet -27

1. Find the derivative of $\sin x^2$ from first principle.

2. Find the derivative of the trigonometric function $\cot\sqrt{x}$ by using chain rule.

3. If $y=\sqrt{\frac{1-\cos x}{1+\cos x}}$, then find $\frac{dy}{dx}$

4. Find derivative function $y=f(x)$, where $y=\sin 2x+(3x+5)^2$

5. Find derivative of $f(x)=x\tan x$

6. Evaluate inverse of trigonometric function $f(x)=\sin^{-1}(x^2)$ from first principle.

7. If $y=\sin^{-1}x^2$, find $\frac{dy}{dx}$

8. Find the second order derivative of trigonometric function $x\sin x$

9. If $y=\sqrt{x+\sqrt{x+\sqrt{x+}}}$....., then find $\frac{dy}{dx}$

10. If 't' is a function of x, then $\frac{d}{dx}\left(\sin^{-1}\right)=\frac{1}{\sqrt{1-t^2}}$